

LIBRARY DATABASE MANAGER

DESCRIPTION:

Library management system is a backend application developed using FastAPI and MongoDB. System tracks records of the staff, students, books and departments in the library and enables the user to update, delete or modify the existing records.

This project is made referencing CRUD operations and relationship mapping. It also provides REST APIs to interact with the database. The APIs are tested using postman without frontend interference.

TOOLS AND PURPOSE

Python: Programming Language
FastAPI: Backend API Framework
MongoDB: Database
PyMongo: MongoDB Connection Library
MongoDB Compass: Database Management
Postman: API Testing
Uvicorn: FastAPI Server

Database Collections

Students: Stores student details and borrowed books.

Books: Stores book information.

Staff: Stores staff details.

Departments: Stores department records.

Conclusion

This project provides practical knowledge of FastAPI, MongoDB, REST APIs, and CRUD operations while demonstrating efficient data handling using a NoSQL database.